

BookFlow – Advanced SQL Operations

JOIN, GROUP BY, Transactions and Indexing

The BookFlow database is extended to manage book loans, generate collection statistics, record book donations using transactions, and improve ISBN-based searching using indexing.

```
DROP DATABASE IF EXISTS bookflow_db;
CREATE DATABASE bookflow_db;
USE bookflow_db;

CREATE TABLE Books (
    book_id INT PRIMARY KEY,
    title VARCHAR(100) NOT NULL,
    isbn VARCHAR(20) UNIQUE,
    published_year INT CHECK (published_year < 2027)
);
INSERT INTO Books (book_id, title, isbn, published_year) VALUES
(1, 'The Great Gatsby', '9780743273565', 1925),
(2, 'To Kill a Mockingbird', '9780061120084', 1960),
(3, '1984', '9780451524935', 1949);
SELECT * FROM Books;

CREATE TABLE Members (
    member_id INT PRIMARY KEY,
    full_name VARCHAR(100),
    email VARCHAR(100) UNIQUE
);
INSERT INTO Members (member_id, full_name, email) VALUES
(101, 'John Smith', 'john.smith@email.com'),
(102, 'Emma Wilson', 'emma.wilson@email.com'),
(103, 'Michael Brown', 'michael.brown@email.com');
SELECT * FROM Members;

CREATE TABLE Loans (
    loan_id INT PRIMARY KEY,
    member_id INT,
    book_id INT,
    loan_date DATE,
    FOREIGN KEY (member_id) REFERENCES Members(member_id),
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FOREIGN KEY (book_id) REFERENCES Books(book_id)
);
INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
(1, 101, 1, '2026-01-10'),
(2, 102, 2, '2026-01-11'),
(3, 103, 3, '2026-01-12'),
(4, 101, 2, '2026-01-13'),
(5, 102, 3, '2026-01-14'),
(6, 103, 1, '2026-01-15'),
(7, 101, 3, '2026-01-16'),
(8, 102, 1, '2026-01-17'),
(9, 103, 2, '2026-01-18'),
(10, 101, 1, '2026-01-19');
SELECT * FROM Loans;
SELECT
    m.full_name AS Member_Name,
    b.title AS Book_Title
FROM Loans l
INNER JOIN Members m ON l.member_id = m.member_id
INNER JOIN Books b ON l.book_id = b.book_id;
SELECT
    published_year,
    COUNT(book_id) AS Total_Books
FROM Books
GROUP BY published_year
ORDER BY published_year;
CREATE TABLE Donation_History (
    donation_id INT PRIMARY KEY,
    book_id INT,
    donor_name VARCHAR(100),

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 donor_name VARCHAR(100),
 donation_date DATE,
 FOREIGN KEY (book_id) REFERENCES Books(book_id)
);
START TRANSACTION;
INSERT INTO Books (book_id, title, isbn, published_year)
VALUES (4, 'Animal Farm', '9780451526342', 1945);
INSERT INTO Donation_History
(donation_id, book_id, donor_name, donation_date)
VALUES
(1, 4, 'Raj Kumar', CURDATE());
COMMIT;
CREATE INDEX idx_books_isbn ON Books(isbn);
SELECT *
FROM Books
WHERE isbn = '9780451524935';
SELECT * FROM Books;
SELECT * FROM Members;
SELECT * FROM Loans;
SELECT * FROM Donation_History;

```

|                                                                                                                                                                                                                                                                              |             |         |            |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|------------|---------------|
| <div> <div>56</div> <div>ORACLE BY published year:</div> </div>                                                                                                                                                                                                              |             |         |            |               |
| <div> <div>Result Grid</div> <div> <div>Filter Rows:</div> <div></div> </div> <div> <div>Edit:</div> <div></div> <div></div> <div></div> </div> <div> <div>Export/Import:</div> <div></div> <div></div> </div> <div> <div>Wrap Cell Content:</div> <div></div> </div> </div> |             |         |            |               |
|                                                                                                                                                                                                                                                                              | donation_id | book_id | donor_name | donation_date |
| ▶                                                                                                                                                                                                                                                                            | 1           | 4       | Raj Kumar  | 2026-10-01    |
| *                                                                                                                                                                                                                                                                            | NULL        | NULL    | NULL       | NULL          |

| Output |             |                                                                                                                                   |                                                          |
|--------|-------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| #      | Time        | Action                                                                                                                            | Message                                                  |
| ✓      | 3 19:52:50  | USE bookflow_db                                                                                                                   | 0 row(s) affected                                        |
| ✓      | 4 19:52:50  | CREATE TABLE Books ( book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) UNIQUE, pu...                         | 0 row(s) affected                                        |
| ✓      | 5 19:52:50  | INSERT INTO Books (book_id,title,isbn,published_year) VALUES (1,'The Great Gatsby','9780743273565',1925), (2,'To Kill a Mo...     | 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0   |
| ✓      | 6 19:52:50  | SELECT * FROM Books LIMIT 0, 1000                                                                                                 | 3 row(s) returned                                        |
| ✓      | 7 19:52:50  | CREATE TABLE Members ( member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE)                              | 0 row(s) affected                                        |
| ✓      | 8 19:52:50  | INSERT INTO Members (member_id,full_name,email) VALUES (101,'John Smith','john.smith@email.com'), (102,'Emma Wilson','e...        | 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0   |
| ✓      | 9 19:52:50  | SELECT * FROM Members LIMIT 0, 1000                                                                                               | 3 row(s) returned                                        |
| ✓      | 10 19:52:50 | CREATE TABLE Loans ( loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIGN KEY (...                        | 0 row(s) affected                                        |
| ✓      | 11 19:52:50 | INSERT INTO Loans (loan_id,member_id,book_id,loan_date) VALUES (1, 101, 1, '2026-01-10'), (2, 102, 2, '2026-01-11'), (3, 103, ... | 10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0 |
| ✓      | 12 19:52:50 | SELECT * FROM Loans LIMIT 0, 1000                                                                                                 | 10 row(s) returned                                       |
| ✓      | 13 19:52:51 | SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.member_id = m.m...                | 10 row(s) returned                                       |
| ✓      | 14 19:52:51 | SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year ORDER BY published_...                    | 3 row(s) returned                                        |
| ✓      | 15 19:52:51 | CREATE TABLE Donation_History ( donation_id INT PRIMARY KEY, book_id INT, donor_name VARCHAR(100), donatio...                     | 0 row(s) affected                                        |
| ✓      | 16 19:52:51 | START TRANSACTION                                                                                                                 | 0 row(s) affected                                        |
| ✓      | 17 19:52:51 | INSERT INTO Books (book_id,title,isbn,published_year) VALUES (4,'Animal Farm','9780451526342',1945)                               | 1 row(s) affected                                        |
| ✓      | 18 19:52:51 | INSERT INTO Donation_History (donation_id,book_id,donor_name,donation_date) VALUES (1,4,'Raj Kumar',CURDATE())                    | 1 row(s) affected                                        |
| ✓      | 19 19:52:51 | COMMIT                                                                                                                            | 0 row(s) affected                                        |
| ✓      | 20 19:52:51 | CREATE INDEX idx_books_isbn ON Books(isbn)                                                                                        | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0   |
| ✓      | 21 19:52:51 | SELECT * FROM Books WHERE isbn = '9780451524935' LIMIT 0, 1000                                                                    | 1 row(s) returned                                        |
| ✓      | 22 19:52:51 | SELECT * FROM Books LIMIT 0, 1000                                                                                                 | 4 row(s) returned                                        |
| ✓      | 23 19:52:51 | SELECT * FROM Members LIMIT 0, 1000                                                                                               | 3 row(s) returned                                        |
| ✓      | 24 19:52:51 | SELECT * FROM Loans LIMIT 0, 1000                                                                                                 | 10 row(s) returned                                       |
| ✓      | 25 19:52:51 | SELECT * FROM Donation_History LIMIT 0, 1000                                                                                      | 1 row(s) returned                                        |